

This file give a very sparse overview of some of the functions we might be able to play with in the workshop for tomorrow (July 17). This is a two-filament spatially-explicit model of muscle contraction. This is very similar to the type of simulation that was run for the Daniel et al. 1998 Biophysical Journal paper:

**Compliant realignment of binding sites in muscle: transient behavior and mechanical tuning**

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All code is written in Matlab. I'm not sure which toolboxes might be needed. This is relatively primitive code and not put together in a very good fashion for anyone who wants to learn how to write proper code. I did however, make quite a few comments that might help if people have questions.

**TwofilMC 04 18 2017.m** Main code file to run.

This main file will then call some other “.m” files as scripts (not functions). Yes—I realize this is not great code etiquette:

Twofilparams.m loads a bunch of variables related to the parameters of the simulation. For example, filament length and stiffness values, the number of embarrassingly parallel runs to loop over to make the Monte Carlo stochastic variability less via averaging, etc.

Build\_mcore.m This builds the structure of the filament lattice (yet only 2 filaments) and will keep track of the mechanical parameters of the system to add in cross-bridge binding and be used to calculate the balance of forces each timestep (with a matrix inversion, that is actually Gaussian elimination, I'm pretty sure).

krates.m This little function calculates the rate functions for a given pair of nodes (one thick-filament node and one thin-filament node), given the positions of the nodes and the state of the cross-bridge. Basically strain-dependent cross-bridge rate functions.

ExpFit.m This takes in whichever data one wants to fit to a single exponential function. This is used at the very end for plotting. This may require the optimization toolbox, and could be commented out or removed if one does not have this toolbox. This function is way more complicated than it needs to be, because at one time I was fitting a bunch of different model outputs across a range of calcium values to look at the force-[Ca<sup>2+</sup>] relationship.